It can be a stirring sight: following an unfair election, thousands of people take to the streets to demand that new, free elections. In some cases, the scale of the protests and solidarity of their participants can precipitate the collapse of a non-democratic regime; the 2004 Orange Revolution succeeded in preventing the theft of a presidential election in Ukraine, for instance, while large protests in Peru helped prevent President Alberto Fujimori from claiming a third term. In other cases, however—despite protesters' determination—incumbents are able to remain in power through a combination of repression and concessions. Large, courageous, but ultimately failed protest movements in Russia in 2011 and Belarus in 2020 offer stark examples.

Successful electoral revolutions figure prominently in theories of election manipulation and how it might be restrained, with broader implications for the stability of both democratic and authoritarian regimes. A classic view, influential in the formal modeling tradition, argues that the use of election manipulation reveals an incumbent's unpopularity, signaling weakness against a post-election challenge. As more extensive manipulation signals deeper weakness, strategic opposition actors should be more likely to initiate protest as election manipulation increases (all else equal). Such models predict that some incumbents, who require manipulation to win an election with high probability, choose instead to gamble on relatively clean elections out of fear of costly protest. This framework has normatively optimistic implications, by showing that stronger opposition movements—those that are more organized, better resourced, and more popular—will be better able to deter incumbents' from holding rigged elections. In turn, this dynamic makes democratic transition and consolidation more likely. Because fraud indicates incumbent weakness in this model, I refer to it as the regime-weakness model of electoral manipulation.

An alternative conceptualization of electoral manipulation has developed in recent years. In this view, developed most prominently by Simpser (2013), election manipulation can be a costly signal of the incumbent's resources and capacities. When incumbents manipulate elections, they demonstrate financial, patronage, and organizational resources; they also demonstrate the ability to bend or flout the law. In this view, incumbents that manipulate elections extensively do not betray weakness in the form of unpopularity; instead they display alternative sources of resilience that can be used to compensate for low popular approval. Moreover, since these signals are costly, they can affect the behavior of other political actors (Gehlbach and Simpser 2015). I call this the regime-strength model.

In this paper, I further develop the theory of a regime-strength model of electoral manipulation, and test it using new empirical approaches. In particular, I argue that extensive electoral manipulation helps induce elite solidarity with the incumbent, if and when post-election protests occur, by two mechanisms. First (and most importantly), recruiting low-level actors to perform illicit electoral manipulation entails significant principle-agent problems. Extensive illicit electoral manipulation is, as a result, a costly signal of incumbents' ability to overcome those principal-agent problems through deep patronage resources and effective control over the security apparatus. Second, this costly indicator of 'strength'—deep pockets and influence over the police, prosecutors, and courts—acts as a coordination mechanism for elites during protest events. During a post-election challenge, the behavior of an individual elite (whether or not to remain loyal to the incumbent) will in part be conditioned on their expectation about *other* elite actors' behavior. The costly signal of resources and capacities produced by extensive illicit manipulation makes it appear more likely that other elites will remain loyal; a potentially self-fulfilling prophecy (Hale 2014). Perversely, and counter to the expectation of much existing theory, this means incumbents have incentives to manipulate *more* heavily when they face greater opposition threat.

The stakes of this debate are significant for theories of democratization and autocratization. The regime-weakness view makes auspicious predictions for democratization and ultimately under-pins the notion of 'self-enforcing' democracy (Fearon 2011). If the latter view holds, the risk of electoral protest may fail to deter election manipulation; greater protest risk may instead incentivize incumbents to manipulate elections more profoundly in order to demonstrate their capacities to other actors in the system. In addition, a sizable literature has been founded on the regime-weakness framework. Scholars working in this mode have argued that manipulation can be kept in check, and ultimately reduced, through the development of institutions that reveal information about election integrity: election monitoring (Daxecker 2012; Hyde and Marinov 2014), more independent courts (Chernykh and Svolik 2015), media activity (Nai, Alessandro 2017), and civil society coordination (Birch and Van Ham 2017).

At first glance, it would seem straightforward to test these competing models by comparing estimates of electoral manipulation with the incidence of post-election protest. The regime-weakness model typically predicts that increasing manipulation (at a constant level of true popularity) increases the likelihood that opposition groups will initiate post-election protest (Beaulieu 2014; Little 2012; Luo and Rozenas 2017; Schedler 2002). By contrast, it might be presumed that the regime-strength model would yield the opposite prediction (Simpser 2013); that protest initiation decreases with the level of manipulation (holding vote-share constant). Yet empirical findings on the relationship between manipulation and protest have been mixed (Ananyev and Poyker 2022; Daxecker 2012; Hafner-Burton, Hyde, and Jablonski 2016; Hyde and Marinov 2014; Lankina and Skovoroda 2017).

There are many possible reasons for these mixed results. One fundamental problem with protest initiation as the dependent variable in this context, however, is the complex and multi-purpose nature of post-election protest. The predictions of the regime-weakness model hinge on the assumption that protest initiation is narrowly instrumental—driven by the expected benefit of overthrowing the incumbent. However, protest initiation may be driven by various actors with different motivations, including citizen grievances and the long-term goals of protest entrepreneurs (e.g. generating publicity, waging factional battles, costly signaling in bargaining for policy change, etc.). Some of these reasons hold even when the likelihood of ousting the incumbent appears remote, meaning that protest can co-occur with signals of incumbent strength. Others of these might confound the apparent relationship between manipulation and protest initiation. A highly unpopular incumbent might need extensive manipulation simply to win the election, but deep unpopularity is also likely to provide fertile ground for protests. In such a scenario, the apparent correlation between high manipulation and a high likelihood of protest could be confounded by citizens' grievances.

This paper does not aim to untangle the thorny relationships among citizen grievance, protest entrepreneurs' motivations, electoral manipulation, and protest initiation. Rather, it seeks to sidestep them by employing novel tests of the regime-strength and weakness models of manipulation. I develop two modeling strategies. First, I estimate propensity scores for post-election protest in a cross-national sample of electoral authoritarian regimes using structural factors that precede the actual level of manipulation in the election. These propensity scores are then used to test how electoral manipulation responds to higher structural protest risk. Second, among the subset of election-years in which protest does occur, I model electoral protest *duration* as a dependent variable. Once protests run their course, the 'strength' of an incumbent (its likelihood of outlasting a protest movement) is no longer latent. Instead, the incumbent's response—repression, concessions, or attrition—and its survival (or not) reveal the incumbents' true strength relative to the opposition. The more quickly a government puts down protests, the greater the underlying (unobserved) distribution of power favors the incumbent over the opposition. By contrast, the longer it takes the government to restore control, the less favorable the distribution of power for the incumbent.

[UPDATE AFTER REVISIONS ->] Using data from 659 non-democratic elections across 100 countries, from 1989 to 2012 (of which 211 experienced protest), this paper finds that governments that manipulate elections more heavily are likely to put down protests more quickly, especially when incumbents claim small vote-shares. This is inconsistent with the regime-weakness model of election manipulation, but consistent with a world in which election manipulation indicates sources of regime resilience. Given these results, it is likely that—while incumbents do not seek to provoke protest—the downside risk for incumbents that engage in election fraud in non-democracies has been overemphasized. This interpretation helps explain several empirical patterns at once: the relative paucity of electoral protests (Brancati 2016), the rarity of their success in overturning incumbent regimes, and governments' attempts to claim implausibly high margins of victory in conjunction with high levels of fraud. At a theoretical level, they also suggest that protest risk should be de-emphasized in research on the causes of election manipulation, in favor of alternative explanations which emphasize the role of front-line actors in delivering manipulation. Finally, they suggest a pessimistic outlook for democratization by election in authoritarian regimes.

[INCORPORATE THIS FROM BOOK ->] This interpretation offers a unified explanation for several secular patterns: the infrequency of electoral protests (Brancati 2016), the limited success of such protests in toppling incumbent regimes, and the tendency of governments to report implausibly large victory margins in heavily rigged elections. For theories of electoral manipulation, it implies that more attention should be given to alternative deterrents to manipulation (such as the role of risk to low-level actors who implement manipulation). Ultimately, this perspective points to a more pessimistic view of democratization through elections in authoritarian contexts, and for the logic of 'self-enforcing democracy.'

1 Literature review: regime strength, weakness, and election manipulation

The literature on authoritarian elections, electoral manipulation, and protest has developed competing (and at times contradictory) assumptions about the nature of regime 'strength' (Leventoğlu, Malesky, and Wen 2025). Fundamentally, a strong regime is one with a high probability of surviving a non-electoral challenge like a mass protest event or an attempted coup (Leventoğlu, Malesky, and Wen 2025). Sources of strength may include the loyalty of the military and security services, patronage resources, the financial or security backing of foreign powers, and more. By surviving—and deterring—challenges, a strong regime is also likely to be *durable*, though even a strong regime can fail.

More contentious is the relationship between the incumbent's popularity and regime strength. The incumbent's popularity is his likelihood of winning a free and fair election.

Some influential models conflate strength with popularity; in these models, popularity determines whether an incumbent is likely to survive a challenge. In those models, electoral manipulation can only signal weakness if exposed, since manufactured votes displace authentic ones for any particular vote total (Leventoğlu, Malesky, and Wen 2025). Popularity is, in my view, another component of strength against extra-electoral challenges. A more popular regime can rely on its genuine supporters to counter-mobilize against protesters or coup plotters (Hellmeier and Weidmann 2020), and popular goodwill can deprive its opponents of recruits (Rosenfeld 2021). Having identified these conceptual distinctions, we now turn to the benefits and risks of election manipulation for incumbents.

1.1 Authoritarian principals and elections

As discussed above, authoritarian leaders face two central threats: mass revolt from below and elite coups (Svolik 2012). Elections help incumbents manage this dilemma by providing information about citizens, opposition groups, and the popularity and effectiveness of subordinates (Blaydes 2011; Geddes and Zaller 1989; Magaloni 2006; Miller 2015), while also helping recruit ambitious politicians and manage intra-elite disputes (Lust-Okar 2006).

Of course, holding an election creates the possibility of *losing* an election. Early research emphasized manipulation as a tool for avoiding electoral defeat. From this perspective, incumbents have more to gain from election rigging when they are likely to narrowly lose the election without it (Birch 2007; Fukumoto and Horiuchi 2011; Kam 2017; Lehoucq and Molina 2002; Nyblade and Reed 2008), and when the prospect of losing carries significant costs (Acemoglu and Robinson 2005; Baturo 2014; Horz 2021; Schedler 2013; Ziblatt 2009). Subsequent research highlighted broader benefits. Excessive or blatant manipulation can signal regime resources and durability, encouraging others to cooperate with the incumbent (Gehlbach and Simpser 2015; Simpser 2013). Costlier forms of manipulation may provide especially credible signals of strength and thereby reduce protest (Harvey and Mukherjee 2020).

On the other hand, manipulation nevertheless entails substantial material and organizational costs. Even relatively cost-effective tactics like co-opting employers to pressure voters requires monitoring efforts and rewards for compliant businesses (Frye, Reuter, and Szakonyi 2014, 207). Agents and brokers engaged in falsification or other types of manipulation must still be organized and compensated (Langston and Morgenstern 2009). Vote-buying can be particularly costly (Lehoucq and Molina 2002; Wang and Kurzman 2007), with costs rising as elections become more competitive (Corstange 2018). Wang and Kurzman (2007), for example, estimate that vote-buying in a single Taiwanese county in 1993 could cost more than $4 million (1993 dollars), roughly twelve times the legal campaign expenditure limit. More broadly, manipulation draws on finite resources—including cash, goods, jobs, regulatory pressure, and favors—that must be traded off against other electoral or political purposes.

Beyond the organizational cost, scholars have also focused on the 'legitimacy cost' of election manipulation (Birch 2011). Legitimacy is understood as an individual's subjective sense that authorities have the right to rule, even if the individual may object to particular decisions (Lipset 1960). Greater legitimacy, in turn, makes it more likely that individuals will comply with authorities' demands (Levi and Sacks 2009). Legitimacy cost, then, is both psychological and potentially behavioral (Schedler 2013). Post-election protest is the most visible, and possibly most consequential, legitimacy cost of electoral manipulation (Birch 2011).

This understanding of electoral manipulation and protest has been widespread in the literature. In several formal models of manipulation, loss of support and/or protest risk often stand as the main downside of manipulation for incumbents (Chernykh and Svolik 2015; Little 2015; Little, Tucker, and LaGatta 2015). Protest risk as the cost of manipulation also figures in less formal theory. It is central to the literature on election monitoring (Hyde and Marinov 2012; Kelley 2012) Manipulated elections can serve as a focal point for citizens' diverse grievances, raising the likelihood of protest (Higashijima 2022; Tucker 2007).

In sum, when holding an election, the existing literature expects incumbents to try to optimize across several goals: to win the election, to send signals of ‘strength’ to other political actors, to manage organizational costs, and to minimize legitimacy costs (typically defined as the risk of protest). As a result, a theory of electoral manipulation must also account for the motivations of post-election protest. Integrating agent risk more fully in the theory of manipulation will show that the most fundamental ‘legitimacy cost’—the risk of losing office in a revolt—actually declines as electoral manipulation increases.

1.2 Post-election protest

Regime-weakness models assume that opposition protest is *elite-led and instrumental*. The opposition is typically modeled as a unitary, strategic actor who stages protest when the expected benefits of successfully overthrowing the incumbent outweighs the expected cost of failure. The signal of weakness sent by election manipulation, in turn, indicates to opposition actors that protest is more likely to succeed in unseating the incumbent (Beaulieu 2014; Little 2012; Luo and Rozenas 2017; Schedler 2002). Incumbents in these models typically face a tradeoff between the short-term benefits of electoral manipulation for winning the election, and any weakness that relying on fraud reveals to the opposition (Rozenas 2016). As Beaulieu (2014, 48) writes in this tradition, “As the opposition grows stronger, the amount of [manipulation] he will tolerate shrinks, and the incumbent who wishes to avoid protest will need to reduce the amount of manipulation she commits.” Such abstraction helps keep formal models tractable, but this assumption is difficult to square with recent research on electoral protest and a broader understanding of political protest in authoritarian regimes.

Electoral protest may occur because manipulation galvanizes citizens who hold diverse grievances (Tucker 2007), because economic conditions are poor (Beaulieu 2014; Brancati 2016), because electoral manipulation makes people angry (Szakonyi 2021), or simply because their party lost the election (Anderson and Mendes 2006). Opposition elites, in turn, might seek to initiate—or amplify—protest to take advantage of this popular discontent for broad, rather than instrumental, reasons. They may wish to pressure the incumbent in order to push for improved competitiveness in the *next* election or to bargain for policy concessions (Chernykh 2014; Robertson 2010). They may stage protests in order to identify new participants, forge new social ties (Barberà and Jackson 2019) and test new tactics. They may seek to communicate information about the government's unpopularity, to set the stage for larger, future protests (Lohmann 1994). Protests may be staged as part of internal factional rivalries within the opposition (Buckles 2017), or to drive media coverage of issues important to the opposition (Huang, Boranbay-Akan, and Huang 2019; Jennings and Saunders 2019). Indeed, opposition elites may take advantage of the bare fact of an election to challenge the incumbent, irrespective of the level of manipulation or the incumbent's vote-share (Pop-Eleches and Robertson 2015). If protest is dispersed and multipurpose, protest movements may initiate for reasons orthogonal to perceived incumbent strength and severity of electoral manipulation. In such a setting, incumbents cannot count on quiet streets simply by manipulating at a low level, or even holding clean elections.

The varied motivations for protest may help account for the inconsistent relationship between election manipulation and protest initiation found in the literature. Several cross-national studies have found links between election manipulation and binary measures of protest (Brancati 2016; Harvey and Mukherjee 2020; Hyde and Marinov 2014). A study of African elections from 1997 to 2009 shows that election fraud is associated with an increased number of post-election conflicts (Daxecker 2012). Similar results are found by Rød (2019), using a different measure of manipulation. Using an alternative measure of manipulation from the NELDA dataset, however, Hafner-Burton et al (2016) find that the association between election fraud and post-election protest falls short of conventional statistical significance. Studies of one archetypal example, the 2011 election and protest cycle in Russia, have found a positive relationship between fraud and protest by some measures (Lankina and Skovoroda 2017), and no relationship by others (Ananyev and Poyker 2022). As noted earlier, Simpser finds a negative relationship between the severity of election manipulation and the likelihood of protest (Simpser 2013) and a positive relationship between 'excessive manipulation' (i.e. extensive manipulation and a large incumbent vote-share) and regime tenure.

[ADD NEW SEGUE TO THEORY SECTION HERE]

2 Theory: electoral manipulation, incumbent strength, and protest duration

BOOK TEXT:

There are two major dimensions to this argument. First, I argue that the same capacities that incumbents rely on to manipulate elections—patronage resources and impunity for agents—are useful for defeating protest movements. Consequently, incumbent regimes that manipulate elections extensively are intrinsically likely to have the tools needed to survive post-election protests. This means that incumbents who are capable of extensive manipulation face reduced legitimacy costs from that same manipulation; they are more likely to survive the possible backlash.

Second, expanding upon Simpser's work (2013) I argue that extensive electoral manipulation sends signals about these resources and capacities to elites, which incentivize them to cooperate with the incumbent during protest movements. Because it requires the resolution of significant principal-agent problems, extensive electoral manipulation is a costly signal of the incumbent's patronage resources, control over the security environment and the courts, and ability to command agent loyalty broadly across the territory of the country. It further signals the regime's intention to devote resources within the elite selectorate, rather than more broadly among society. The more extensive the manipulation, the stronger these signals in favor of elite solidarity are likely to be.

This model extends Simpser's work in two ways. First, it grounds the costly nature of the strength signal in the principal-agent dynamic; if manipulation could simply be willed into existence by the incumbent, it would not be costly (Harvey and Mukherjee 2020). By overcoming a huge set of principal-agent problems, the incumbent 'pays' to display his patronage capacity and influence over agents. This is a fundamental reason why electoral manipulation is not cheap talk, as some deterrence models posit. Second, I argue that these signals do not require manipulation to be extensive or blatant, as in Simpser (2013). To be successful in this regard, manipulation need not result in a wide margin of victory or be carried out in plain view; it is only necessary that the manipulation effort be intelligible to elites themselves.

2.1 Electoral manipulation, principal-agent problems, and costly signaling

Extensive electoral manipulation requires cooperation from a large network of subordinates who bribe or pressure voters, stuff ballot boxes, or falsify results (Rundlett and Svolik 2016). These agents may include partisan activists (Stokes, Dunning, and Nazareno 2013), election officials (Bader 2017; Sjoberg 2016), neighborhood brokers (Langston and Cornejo 2023), local mayors and civil servants (Mares and Young 2018), public employees (Forrat 2018), and workplace managers (Cohn-Postar 2021; Frye, Reuter, and Szakonyi 2025).

Mobilizing thousands of such individuals poses substantial principal-agent problems. Whereas early research often assumed that incumbents' orders were smoothly implemented, subsequent research has emphasized the risks posed by subordinates (Magaloni 2008; Svolik 2009; Svolik 2012), pervasive principal-agent problems (Harvey 2019; Rundlett and Svolik 2016; Tyson 2018), and difficulties monitoring low-level agents (Boix and Svolik 2013; Larreguy, Marshall, and Querubin 2016a). These networks are typically pyramidal, with each level overseeing a larger number of actors below it (Hale 2014). Yet agents may not behave as principals intend (Rundlett and Svolik 2016), and actors throughout the hierarchy have incentives to misrepresent their loyalty and effort (Harvey 2022b). Large-scale manipulation therefore depends on a hierarchical network in which high-level actors such as party leaders, governors, and business owners influence meso-level actors such as mayors and firm directors, who in turn direct large numbers of front-line agents such as election officials, civil servants, and party operatives.

Agents' incentives also diverge fundamentally from those of political principals. Principals benefit directly from manipulation because it increases their chances of winning close elections (Lehoucq and Molina 2002) and can widen their margins of victory (Gehlbach and Simpser 2015; Greene 2007; Magaloni 2006; Simpser 2013). Agents, by contrast, are primarily motivated by access to patronage—jobs, rents, and other benefits associated with political networks (Diaz-Cayeros 2006; Lust-Okar 2006; Reuter and Robertson 2012)—and by the risk of exposure and punishment for breaking the law (Harvey 2022a).

Consequently, incumbents who produce extensive electoral manipulation must have deep patronage resources (Greene 2007; Hale 2014; Harvey 2019) and the ability to shield agents from the consequences of exposure. By overcoming a large number of nested, geographically dispersed principal-agent problems, an incumbent has produced a costly signal of those resources and capacities. Crucially, many of the same resources and capacities improve the incumbent's ability to respond to a nascent protest movement through repression or concessions (Bishara 2015; Frantz and Kendall-Taylor 2014; Hummel 2019; Lipsky 1968; Rasler 1996; Tilly 1978). Patronage resources may be aimed at protesting parties themselves (Hafner-Burton, Hyde, and Jablonski 2016; Moore 2000; Piven and Cloward 1979; Reuter and Robertson 2015; Robertson 2010) or the more moderate public, in order to keep them on the sidelines of the conflict (Hummel 2019). Incumbents may also threaten to withhold resources from clients in exchange for passive support (Rosenfeld 2017), sapping the pool of potential protesters.

Ruling parties may also lean on repression, calling upon the security services to disperse protests, arrest leaders, and perhaps employ harsher methods including disappearance, torture, and execution (Davenport 2007; Escribà-Folch 2013; Vogel 2022). The incumbent's ability to devote patronage resources to the security services helps maintain their loyalty during such moments, by suggesting that the security forces have more to lose if the incumbent is ousted (Albrecht and Ohl 2016; Bellin 2004). The ability to protect regime agents from any legal consequences for carrying out orders, a condition which helps enable illegal election manipulation (Harvey 2022a), is also an asset when asking law enforcement agents to carry out possibly violent repression (Abouharb, Moyer, and Schmidt 2013; Magaloni and Rodriguez 2020; Stavro and Welch 2023).

This is "infrastructural power" (Leventoğlu, Malesky, and Wen 2025)—those sources of regime durability that derive from its ability to mobilize and discipline agents, to distribute patronage, and to deploy coercion. This contrasts with the legitimacy-based view of strength centered in the regime-weakness framework. As a result, I expect that large-scale electoral manipulation will positively influence elites' expectations of the regime's long-term durability, helping prevent elite splits during a post-election challenge (Bueno de Mesquita et al. 2005; Svolik 2012).

As articulated by Simpser (2013), electoral manipulation can communicate information about the resources and capacities of the incumbent ruling party. This information can then persuade other political actors to remain loyal to the incumbent, rather than join a challenge. Incorporating the principal-agent dynamics discussed in this book shows why electoral manipulation need not be excessive or blatant to serve this function. In Simpser's model, other political actors learn about the level of electoral manipulation through its public display and/or by observing an implausibly high margin of victory for the ruling party. The model articulated here shows how manipulation efforts require the cooperation of large numbers of people, including elites within the regime.[1] Internal elites observe their own level of manipulation effort, and can estimate those of their peers (Rundlett and Svolik 2016). An extensive electoral manipulation effort can therefore be covert to the public, while intelligible to elites. Knowing the difficulty of mobilizing agents to engage in illicit manipulation, such an effort signals regime strength and should prevent splits from within the elite.

[1]Regional governors, big business owners, and directors of security forces are all examples of elites whose resources can be mobilized at election time.

2.2 Regime strength and protest duration

Considering how electoral manipulation may be tied to protest duration under the regime-weakness and regime-strength models reveals distinct predictions. Specifically, when manipulation increases, protest movements should last longer under the regime-weakness model, but be defeated more quickly under the regime-strength model (conditional on incumbent vote-share).[2] The logic hinges on the assumption that, on average, authoritarian governments will seek to quash protest movements quickly once they appear. How plausible is this assumption?

Mass protest is one of the two primary threats to authoritarian regimes (Svolik 2012), and as such they often go to great lengths to prevent it. These efforts include costly (and risky) investments in the repressive apparatus (Dragu and Przeworski 2019; Escribà-Folch 2013), co-optation of opposition parties (Reuter and Robertson 2015), and large-scale monitoring and censorship schemes (King, Pan, and Roberts 2013). And they often relentlessly hound would-be protest entrepreneurs to deter such behavior (Mehrl and Choulis 2024; Xu 2021). Mass protest then, is clearly undesirable from the authoritarian incumbent's point of view, and we can plausibly assume that once an electoral protest movement begins, the incumbent will exert considerable effort to quash it. Governments that are stronger, relative to the protesting opposition, will defeat protest movements faster than will weaker governments. As argued below, electoral manipulation can be demonstrative of capacities that may also be used to defeat protest movements.

An important objection could be raised here. Might not a stronger incumbent, feeling secure in their power, be willing to tolerate opposition protest longer than a weak incumbent? It is true that incumbents do sometimes tolerate protest movements (Yuen and Cheng 2017). However, there are two reasons this is unlikely to be the case systematically here. First, authoritarian governments are more likely to tolerate protests aimed at low-salience, local

[2] In both cases, this assume that incumbent survives the protest movement, by far the modal outcome. When the opposition wins, an outcome considered in the appendix, the assumption would be reversed: a quicker opposition victory indicates a weaker incumbent.

issues (Klein and Regan 2018; Lorentzen 2013), and more likely to clamp down on protests that have higher concession costs (Klein and Regan 2018). As electoral protests challenge the incumbent's ostensible victory and right to remain in office, they are directly threatening to the regime; the concession costs of stepping down or holding new elections are very high. Second, authoritarian leaders are working in an information-poor environment (Guriev and Treisman 2020), in which mass protests can unexpectedly grow from small sparks to wildfires that quickly sweep away the regime (Kuran 1991). "Ignoring" protests can lead to further mobilization against authoritarian regimes (Bishara 2015), as can concessions in the absence of durable political reform (Leuschner and Hellmeier 2024). In such cases, we should expect to see apparent 'toleration' displaced by 'attrition'—efforts by the incumbent to increase the cost of protest while avoiding open repression (Yuen and Cheng 2017). As a control, however, I include a measure of average protest size in one model of protest duration, in the expectation that larger protests are harder to tolerate.

Under what circumstances should protest movements be expected to last longer in authoritarian regimes? Authoritarian incumbents have strong incentives to keep protests small and short-lived. Generally speaking, by definition, authoritarian regimes have a smaller winning coalition and selectorate than democracies do, and allocate more private and club goods to elites as a result (Bueno de Mesquita, Smith, Siverson, and Morrow 2005). Despite this, they can appear superficially popular, with most citizens falsifying their preferences and demonstrating support for the regime, tacitly or otherwise. The appearance of even a moderate number of protesters publicly signalling their disapproval of the government can trigger a cascade of dissent in short order, as other citizens stop falsifying their preferences as well (Kuran 1991). Given time, persistent protests can lead elites and the security services to update their prior beliefs about the strength of the incumbent, prompting defections (Chenoweth and Stephan 2011; Svolik 2012). Protest movements can also challenge the regime's legitimacy (Keremoğlu, Hellmeier, and Weidmann 2022), eroding individual compliance in other areas. To prevent all these negative outcomes, incumbents try to prevent mass protest from occurring, and to quickly quell those protest movements that do arise.

Assuming the incumbent ultimately remains in office, stronger protest movements will tend to last longer than weaker ones. Ordinary protesters are more likely to remain engaged when they are more aggrieved (Gurr 2015), when they are angrier, and when they observe many other people expressing their discontent; extensive public grievance can thus be a source of durability for protest movements. The longer-lasting and larger protest movements are, the more protesters may become socially networked, enabling easier subsequent mobilization (Schussman and Soule 2005; Stryker 2000), increasing organizational resilience and capacity . When group goals appear more achievable—in part because the protest movement is large, well-organized, and well-resourced, and the incumbent appears off balance—individuals we will be more willing to participate in protest (Klandermans 1997) and to become regulars (Saunders et al. 2012). Protest entrepreneurs, who are using protest as a form of costly bargaining against the incumbent, will aim to demonstrate the resilience and popularity of their movements in order to present a stronger negotiating position (Beaulieu 2014; Robertson 2010). As a result, they often seek to maintain high levels of mobilization even in the face of repression and ordinary citizens' waning interest over time (Tilly and Tarrow 2015). On average, then, I expect that authoritarian and proto-authoritarian incumbents will seek to demobilize protest, and protest movement leaders will seek to maintain (or even increase) high levels of mobilization. Which side prevails, and how quickly, depends on the relative strength of each.

Electoral manipulation has been shown to be more extensive under conditions that minimize that minimize principal-agent problems (i.e. those that make low-level agents more likely to cooperate with the incumbent). These include: greater local popularity for the incumbent (Rundlett and Svolik 2016), greater incumbent control over patronage resources (Hale 2014; Harvey 2019), and lower risk of punishment for pro-regime lawbreaking (Harvey 2022a). As discussed below, these capacities and resources are also independently useful for surviving post-election protest. Consequently, if an incumbent is able to manufacture a large numbers of votes, it is also likely to have the resource capacity, control over bureaucratic institutions, and perhaps deep wells of popularity in parts of the country that can serve as independent sources of regime durability, even if the incumbent's nationwide margin of victory is low. Lastly, as I have argued in this volume, by overcoming principal-agent problems arising from agent risk, extensive illicit manipulation also indicates that large number of agents assess that the incumbent will remain in power in the future (Simpser 2013). This can itself become a self-reinforcing equilibrium (Hale 2014).

Still, a full accounting of a regime's strength would include the incumbent's "infrastructural power" *and* their genuine popularity. Like infrastructural power, real popularity among some constituencies should help the government survive a post-election challenge (Gerschewski 2013). Of course, in most authoritarian cases, the incumbent's true popularity is unknown due to censorship, repression of opposition candidates, and preference falsification by citizens. As a result, scholars (and political actors in real time) are left to use the incumbent's vote-share and perceived level of manipulation as imperfect indicators of the incumbent's true popularity. For example, an election in which the incumbent records a high vote-share with limited electoral manipulation might be relatively popular while lacking the capacity to manipulate elections extensively.[3] If a post-election challenge were to occur here, the incumbent's low infrastructural power (e.g. limited patronage resources and weak control of the security forces) might be offset by its relatively high popularity, enabling the government to survive the challenge. Likewise, a low incumbent vote-share in a highly rigged election indicates low popularity but high infrastructural power; the risk of losing a challenge due to low popularity may be offset by the incumbent's other resources and capacities. Furthermore, even genuinely popular incumbents may resort to large-scale manipulation in order to bolster turnout (Martinez i Coma and Morgenbesser 2020; Smyth and Turovsky 2018). Since genuine popularity cannot be directly measured, but is indicated by the mix of incumbent vote-share and level of manipulation, both indicators must be included in our hypotheses.

[3] Such a case might also indicate an incumbent with high capacity to manipulate who *chooses* not to do so, a prospect considered in the next section.

Putting it all together: this theory assumes that authoritarian incumbents generally prefer order to unrest, and that protest leaders want to demonstrate the popularity of their cause through large, sustained protests. Electoral protests are a form of competition between the incumbent and opposition actors, in which both sides are trying to demonstrate their strength in order to win concessions from—or defeat outright—the other (Beaulieu 2014). The outcome of this contest reveals the underlying strengths of the participants. Consequently, governments that quickly defeat protest movements can be considered stronger than those that take weeks or months to prevail, which in turn can be considered stronger than those governments which collapse in the face of mass protest. Incumbents who manipulate extensively demonstrate they have resources and capacities that can also be used to buy off, repress, or otherwise quash post-election protest; they also send costly signals of such to elites that can induce their loyalty. Such signals of extra-electoral strength should be especially informative to elites when regime popularity is low. This yields Hypothesis 1.

> H1: Election manipulation is negatively associated with protest movement survival, with the size of this effect decreasing with incumbent vote-share.

[ALTERNATIVE VERSION:]

> **H1 (Unpopularity offset by instractural power)**: Increases in electoral manipulation will be associated with decreasing duration of post-election protest when incumbent vote-share is low.

All told, the core of the argument is as follows: incumbents generally aim to manipulate elections as extensively as possible to reap both the direct and indirect benefits of doing so (Simpser 2013). Achieving this requires overcoming principal-agent problems—by tightening control over patronage networks and insulating agents from potential consequences. The very resources that enable large-scale manipulation—such as genuine popularity, control over state resources, and dominance of the legal system—also equip incumbents to suppress or dismantle protest movements when they arise. Consequently, more extensive manipulation should correlate with shorter durations of anti-incumbent protests.

This argument—that successful, large-scale election manipulation draws on the same resources a state may use to suppress protest—turns the deterrent logic on its head. Instead of signalling weakness, large-scale fraud indicates a well of resilience: control over resources, impunity for pro-regime lawbreaking, and loyal agents. As a result, governments that build and deploy an electoral manipulation apparatus are simultaneously inoculating themselves against the expected cost of protest.

[OLD TEXT:]

The central argument of this manuscript is that, in authoritarian regimes, incumbents who develop the tools necessary to manipulate elections extensively will also have the tools necessary to more quickly prevail over protest movements. This proposition builds off Simpser's foundational argument that manipulated elections can send costly signals of the incumbent's "capacities, resources, [and] dispositions" that will help it retain power in the future (Simpser 2013, 80)—that is, of its political strength. However, I depart from Simpser in two ways. First, by incorporating more recent research on the principal-agent dynamics inherent in election manipulation, I argue that incumbents who can produce large amounts of election manipulation necessarily have "capacities, resources and dispositions" helpful for defeating protests—a peripheral subject for Simpser. Second, mine is not a signalling argument with regard to protesters (though the election outcome may still communicate signals of regime strength to, for instance, state agents deciding how to respond to the protest movement). I argue that electoral protest may occur for many reasons, even when incumbents are perceived to be 'strong.' However, good manipulators are likely to have deeper wells of strength than ineffective manipulators do, even when they fail to 'deter' post-election protest. This is due, in part, to the regime resources and cohesiveness that widespread election manipulation requires, and in part due to the fact that widespread manipulation is a costly signal of such capacities.

It is likely that the theoretical mechanisms described above weaken as regimes become more liberal-democratic, as Simpser notes (2013, 192); in liberal democracies, patronage consolidation should decline and rule of law should increase by definition. These changes should make it more difficult for incumbents to manipulate elections, irrespective of protest risk. At the same time, the more liberal-democratic a state becomes, the more alternatives there are to protest as a means to check the executive. More independent courts and more active executive constraints have been shown to be associated with improved election integrity (Bishop and Hoeffler 2016; Harvey 2022a); it is where these factors are weakest—the electoral authoritarian regimes—where we should expect the deterrence logic to operate under a regime-weakness model. Finally, as a matter of measurement, protest duration is likely to be an increasingly imperfect measure of incumbent strength in more liberal-democratic settings, where protest is more commonplace and less immediately threatening to the regime.

This model claims that more extensive illicit manipulation serves as a costly signal or regime strength that helps induce elite loyalty during post-election challenges. As such, it also makes a distinctive prediction about how incumbents should respond to the possibility of post-election contestation. Rather than back down in the face of protest risk, as regime-weakness models predict, the regime-strength model suggests that incumbents should work to intensify manipulation. By doing so, they can send a more convincing signal of their staying power to elites, preventing defections, and ultimately *reducing* the risk that the incumbent will lose office.

> *H2 (Signaling to induce loyalty)*: Electoral manipulation should increase with pre-election indicators of protest risk.

In other words, the theory predicts that *incubments should manipulate more when they face strong oppositions*, rather than moderate their behavior, as conventional wisdom might hold, and that when post-election protests do occur, *greater electoral manipulation will be associated with quicker defeat of protest movements.*

3 Data, case selection, and main variables

These hypotheses are tested using data from three datasets: NELDA (Hyde and Marinov 2012), V-Dem version 11.1 (Coppedge et al. 2021), and the Electoral Contention and Violence dataset (Daxecker, Amicarelli, and Jung 2019). The unit of observation for this study is an election period as recorded in NELDA. This prevents selecting on the dependent variable, by capturing both protested and non-protested elections. I limit the sample to authoritarian regimes which hold multiparty national elections as coded in V-Dem. This excludes fully closed autocracies (where no national elections are held), as well as electoral and liberal democracies. I limit the sample to those regime-years coded 1 (electoral authoritarian) on the V-Dem variable 'v2x_regime' for the election year, or coded 2 (electoral democracy) for the election year but 1 for the lagged year.[4] This sample includes a range of electoral authoritarian cases, including those that might be labeled as 'hegemonic' and 'competitive' (Donno 2013).[5] For instance, the dataset includes election years from Egypt and Belarus, but also from Ukraine, and Mexico in the 1990s.

The case selection strategy here, excluding democracies, is one that both fits the theory and makes for a stronger test of the regime-weakness model. Nevertheless, an expanded sample that includes a set of electoral democracies provides similar results in the appendix. In total, there are 659 elections included in the dataset, across 100 countries, with 211 experiencing post-election protest. The date range is limited by the availability of data on protest from ECAV, which ranges from 1989 to 2012.

To capture the severity of *election fraud*, I use the V-Dem measure of intentional election irregularities, a category which includes "double IDs, intentional lack of voting materials,

[4]This coding rule will capture cases that are coded as electoral democratic after an election but electoral authoritarian before, in order to ensure that cases which experience a liberalization after the election are included.

[5]According to the V-Dem codebook, the coding scheme includes cases where multiparty elections were manipulated to the extent that "irregularities... affected the outcome of the election," and ones in which it is "hard to determine whether the irregularities affected the outcome or not." Only when irregularities definitively did not affect the outcome would election-years be categorized as electoral democracy, and excluded from the main analysis.

ballot-stuffing, misreporting of votes, and false collation of votes." It is important to note that this measure captures illegal election-day manipulation broadly, and is not confined simply to falsification.

I operationalize electoral manipulation using this variable, rather than alternative measures of election integrity, because fraud of this kind is thought to be most likely to trigger protest (Linebarger and Salehyan 2020)—it is often incumbents' tool of last resort (Sjoberg 2016), may not mobilize real voters (Harvey and Mukherjee 2020), and can be more damaging to legitimacy than pre-election manipulation (Birch 2011; Szakonyi 2021). This measure of election integrity should thus serve as a strong test of the regime-weakness model: if manipulation indicates incumbent weakness, it will be most likely to do so when employed in this way. It also matches the conception of election manipulation used in much of the relevant formal literature.[6]

Data on protest occurrence and duration are derived from ECAV, which records protests and other events related to an election, ranging from six months before to three months after the vote (or three months after the final round of a multi-round election). To qualify, an action must be contentious; rallies in support of the incumbent are excluded. I only include events labeled as protests, occupations, or blockades, excluding acts like shootings, bombings, and coups.

4 Analysis 1: Protest risk and severity of manipulation

4.1 Modeling strategy

I utilize two complementary approaches to test the relationship between protest risk and manipulation. The first of these takes the *level* of election manipulation as the outcome variable, while the second uses *changes* in the level of manipulation. Both employ two-stage

[6] For example, election manipulation is treated as post-hoc adjustment of the results in Little et al (2015), Little (2012), Magaloni (2010), and Fearon (2011), among others.

models.

In the first approach (where level of manipulation is the ultimate outcome variable), a logit model is used in the first stage to generate a propensity score of post-election protest in each case; the first-stage model uses only pre-election variables as predictors. The goal of this step is to estimate the underlying likelihood of post-election protest in each country-year, independent of the level of election fraud observed on election day. The fitted values (i.e. predicted probabilities of protest) for each observation are obtained as propensity scores for protest, to be used as explanatory variables in the second-stage models. The second-stage models take the level of *electoral fraud* as dependent variables. Deterrence models would expect that a higher propensity for protest would be associated with lower levels of manipulation, while the 'regime-strength' approach predicts the opposite; incumbents should work to intensify manipulation when protest risk is higher in order to signal strength to elites.

It is important to clarify that I am not implementing a propensity-score *matching* or weighting approach here. Actual post-election protest cannot be considered a cause of electoral manipulation which occurs *before* the protest, and thus cannot be operationalized as a treatment variable. Instead, I aim to estimate the latent protest risk in each case, and analyze that risk as a predictor of observed electoral manipulation.

Stage one predictors The effectiveness of this propensity score approach hinges on the success of the first-stage model, which should include variables that influence the likelihood of treatment. The first-stage model here includes lagged *judicial independence* (Linzer and Staton 2015), which can influence parties' decision to challenge elections in the streets vs. the courtroom. It also includes several lagged control variables taken from V-Dem, including measures of civil society organization ('v2cseeorgs') and legislative constraints on the executive ('v2xlg_legcon'), as estimates of the degree of contestation in the political system. The lagged openness of the media environment ('v2x_freexp_altinf') is included, since this could influence citizens' ability to gather information about the quality of elections and to communicate their grievances. De facto protection of individual physical integrity is included (lagged) along with its square term ('v2x_clphy'), to account for the linear and curvilinear effects of repression on protest. Lagged GDP growth ('e_migdpgro') is included as a measure of economic grievances. Finally, data on urban population share (from the World Bank) is included, since population density is a common correlate of protest.

Three binary variables capture relevant features of the election, taken from NELDA. *Presidential* elections may have higher stakes, and thus be more prone to sparking protest. *Off-schedule elections* may be strategically timed by incumbents to account for more elevated protest risk. In addition, whether the executive is facing *term limits* is also included, since this can lead to fragmentation in the elite that can in turn yield protest Robertson (2010). The occurrence of a successful regional electoral protest in the previous year is included, derived from the ECAV data, given the potential to inspire protests in neighboring countries.

Lastly, two final—and very important—covariates are included. These are: the number of pre-election protests (a very direct measure of protest risk), and lagged overall election integrity (i.e. overall manipulation in the previous national election, 'v2elfrfair' from V-Dem).

The results of the first stage model (with protest initiation as the DV) are presented in Table 1. This model appears to capture the latent risk of protest reasonably well; the average propensity score for country-years where no protest ultimately occurred is 0.12. By contrast, the average propensity score for country-years where protests *did* occur is 0.66.

In the second approach, I first model the average level of manipulation for each country, using only structural variables unrelated to protest risk. The residuals from this model—the observed deviation from the average for each country-year—can be interpreted as a measure of over- or under-production of manipulation relative to each country's baseline. I regress these residuals on of pre-election predictors of protest. Where the first approach estimates the level of fraud, the second approach estimates change in levels over time. As before, a deterrence model would predict that higher protest risk is negatively associated with these residuals (i.e. with *under*-production of illicit manipulation compared to baseline), while the regime-strength approach predicts the opposite.

Table 1: Stage 1 models

	Model 1	Model 2
	Protest (logit)	Fraud (OLS)
(Intercept)	−2.587**	0.665*
	(0.917)	(0.321)
Presidential election	1.257***	0.005
	(0.297)	(0.040)
Judical independence (lag)	−0.145	−1.023**
	(1.247)	(0.361)
Leg. constraints (lag)	0.768	
	(0.814)	
GDP growth rate (lag)	−0.009	0.149*
	(0.542)	(0.061)
Alternative info. (lag)	−0.662	
	(1.238)	
Civil soc. openness (lag)	0.446+	
	(0.243)	
Incumbent term-limited	−1.299+	
	(0.688)	
Urbanization	0.005	0.002
	(0.009)	(0.007)
Overall elec. integrity (lag)	−0.250	
	(0.971)	
Number of pre-election protests	19.298	
	(674.968)	
Phys. integrity index (lag)	1.534	1.758**
	(3.198)	(0.627)
Phys. int. lag squared	−2.189	−1.477*
	(3.096)	(0.613)
Regional protest diffusion	0.503	
	(0.382)	
Country FE	Y	N
Num.Obs.	527	600
R2		0.826
R2 Adj.		0.791
AIC	361.6	842.8
F	1.986	
RMSE	0.31	0.41

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Stage 1 predictors The stage one, OLS model uses only structural variables to predict baseline manipulation. Country fixed effects are included so that this baseline level of fraud is isolated by country. Factors associated with classic protest deterrence, like the number of pre-election protests, are excluded. This should produce an estimate of the structural level of manipulation the incumbent could produce in the absence of strategic deterrence. The predictors include a V-Dem measure of the autonomy of election management bodies ('v2elembaut'), since less autonomous EMBs could be associated with higher levels of manipulation. A similar logic applies to the professionalization of the national bureaucracy ('v2stcritrecadm') and the degree of state control over the economy ('v2clstown'). Since access to and the source of government resources affect manipulation capacity, I include categorical variables for states that are primarily financed through taxation or resource rents ('v2stfisccap').[7] The model also includes several predictors from the previous analysis: lagged judicial independence, lagged GDP growth rate, whether it is a presidential election, the degree of urbanization, and lagged physical integrity and its square term.

Stage 2 predictors

4.2 Analysis 1 Results

Table 2 shows the results of the second-stage models from both approaches. Model 3 shows that an increase in the pre-election propensity score of protest is associated with a sizable and statistically significant *increase* in subsequent manipulation. Model 4 shows that a recent successful protest movement in the region is also associated with more severe election fraud, while controlling for a variety of possible confounders. Model 5 includes all controls, the regional protest dummy, and the protest propensity score; the regional protest dummy remains positive and significant at the conventional level. Finally, Model 5 shows that increasing pre-election protests and successful electoral protest in the region are positively associated with increased fraud *relative to each country's baseline*. In other words, increasing

[7] The omitted, baseline category in the regression model captures weak states with little domestic access to revenue.

Table 2: Stage 2 models of election fraud (OLS)

	Model 3	Model 4	Model 5	Model 6
(Intercept)	0.982***	1.677***	1.705***	−0.026
	(0.060)	(0.194)	(0.205)	(0.019)
Judical independence (lag)		−2.609***	−2.469***	
		(0.270)	(0.283)	
Leg. constraints (lag)		−0.898***	−0.930***	
		(0.179)	(0.192)	
GDP growth rate (lag)		0.077	0.073	
		(0.090)	(0.091)	
Alternative info. (lag)		0.832**	0.692*	
		(0.291)	(0.306)	
Civil soc. openness (lag)		−0.083	−0.035	
		(0.054)	(0.060)	
Presidential election		−0.085	−0.101	
		(0.072)	(0.077)	
Incumbent term-limited		−0.013	−0.041	
		(0.138)	(0.142)	
Urbanization		−0.004*	−0.004+	
		(0.002)	(0.002)	
Off-schedule election		−0.288***	−0.248***	
		(0.068)	(0.072)	
Number of pre-election protests		0.006+	0.002	0.004*
		(0.003)	(0.004)	(0.002)
Phys. integrity index (lag)		3.071***	2.755***	
		(0.670)	(0.713)	
Phys. int. lag squared		−3.917***	−3.689***	
		(0.663)	(0.693)	
Regional protest diffusion		0.211*	0.224*	0.110*
		(0.089)	(0.097)	(0.046)
Protest risk	0.268*		0.186	
	(0.119)		(0.116)	
Num.Obs.	527	561	527	600
R2	0.010	0.393	0.393	0.017
R2 Adj.	0.008	0.379	0.376	0.014
AIC	1484.2	1322.8	1252.2	636.4
F	5.038	27.294	23.665	5.231
RMSE	0.98	0.77	0.77	0.41

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

protest risk is not only associated with higher levels of fraud cross-nationally, but also with intensifying fraud *within countries*.

Where the regime-weakness model predicts that protest risk should deter manipulation, the results find just the opposite: incumbents appear to manipulate more heavily under riskier conditions. This is a theoretically significant result, and is also evidence in favor of the signaling mechanism described in the theory section; as the risk of post-election protest rises, incumbents are incentivized to manipulate more heavily to signal their resources and capacities to within-system elites.

5 Analysis 2: Severity of manipulation and protest duration

The main variables used in this analysis are a count measure of *protest duration* in days, and an estimate of the severity of *election fraud*. These are discussed here, with control variables discussed later.

Data and modeling strategy ECAV data is in event-day format; to capture *protest duration* I take the number of days between the last protest recorded in ECAV and the first, for each election.[8] Protest wave duration is modeled using Cox proportional hazards models, which have the following formula:

$$h(t|X) = h_0(t) * exp(\beta_f X_f + \beta_v X_v + ... + \beta_p X p_)$$

The formula models the risk that a protest movement will fail at time t given covariates X is a function of the baseline hazard function, the degree of electoral fraud (X_f), the incumbent's vote-share (X_v) and a set of control variables. For the survival models, protest movements that were ultimately defeated by the incumbent are considered 'dead' after the last day of protest. Protest movements that prevailed in unseating the incumbent are included, and are

[8] The duration of a protest wave with only one protest is taken to be one day.

marked as ‘alive.’

The large majority of elections experience no post-election protest: in 448 of the 659 elections, no protests are recorded. Figure 1 reports the distribution of election fraud across the dataset, as well as protest duration. As the figure shows, while election fraud is common among this set of regimes, protests are rare, and, when they do occur, are usually short in duration. These distributions—extensive election fraud combined with large numbers of non-protest—provides some early indication that the regime-weakness model has limited support. Still, approximately one-quarter of elections experienced some protest, and so protest initiation is modeled using standard logistic regression. Protest wave duration is modeled using Cox proportional hazards models.

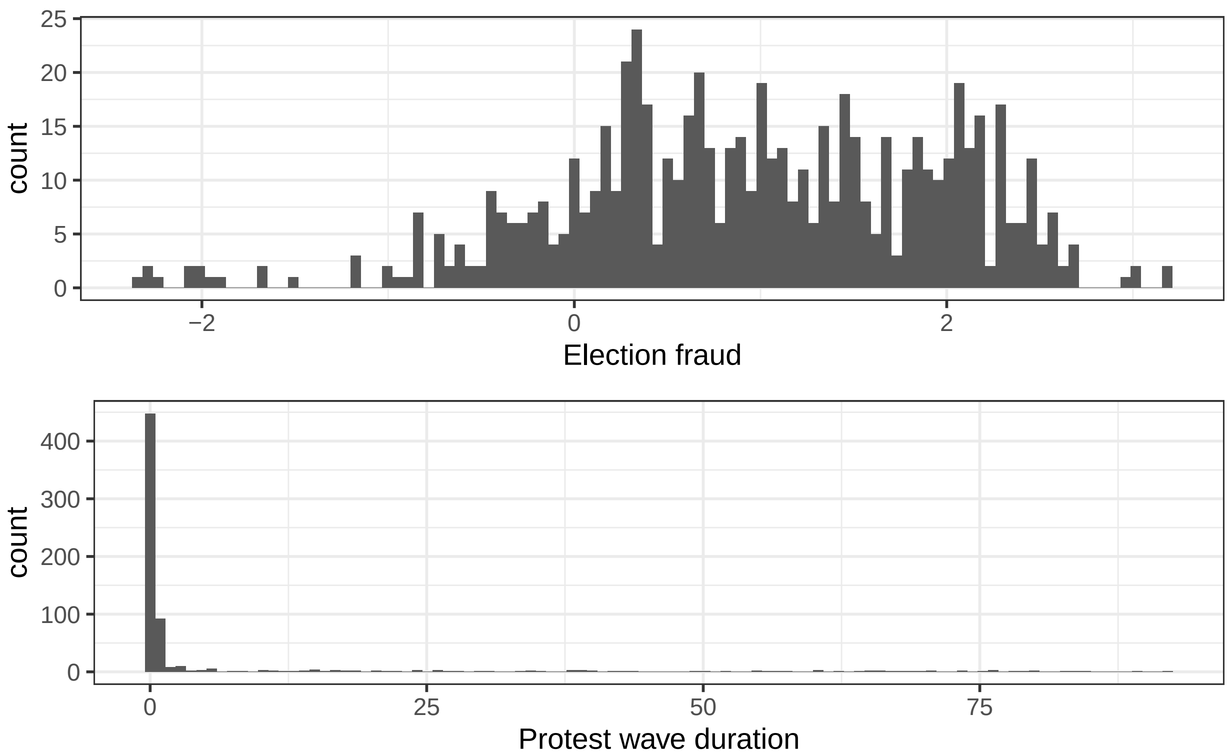

Figure 1: Histograms of election fraud and protest duration

Explanatory variables:

Second, the *incumbent party’s vote-share* is also taken from the V-Dem dataset.[9] The

[9]This variable has considerable missingness, which I reduced in part by updating missing values in V-Dem using data from Wikipedia, where links to official sources were available, and Psephos.

original V-Dem variable indicates the *winner's* vote-share. However, I exclude all cases where the incumbent lost the election outright. This limits the observations to those that fit the population of interest: protest following an election in which the incumbent claimed victory, possibly through reliance on election fraud; it also ensures that the vote-share variable captures the incumbent's position.

Excluding cases where incumbents stepped down after losing an election avoids introducing measurement error in the dependent variable, due to a quirk of the ECAV dataset, which codes the 'side' of the protest (pro- or anti-government) as missing during the period between an election and the swearing in of a new government in cases where the incumbent loses. Including these cases thus runs the risk of conflating pro- and anti-incumbent protest, along with cases where protesters goals are genuinely unknown. However, excluding them also influences the interpretation of the results from these models. Threats to inference from this section are evaluated in more detail in the discussion section.

For the models of protest duration, I exclude all cases where the incumbent lost the election outright. This limits the observations to those that fit the population of interest: protest following an election in which the incumbent claimed victory, possibly through reliance on election fraud. Excluding cases where incumbents stepped down after losing an election avoids introducing measurement error in the dependent variable, due to a quirk of the ECAV dataset, which codes the 'side' of the protest (pro- or anti-government) as missing during the period between an election and the swearing in of a new government in cases where the incumbent loses. Including these cases thus runs the risk of conflating pro- and anti-incumbent protest, along with cases where protesters goals are genuinely unknown. However, excluding them also introduces potential selection effects into the models. Multiple additional analyses in this chapter indicate that bias from selection effects is not present; moreover, they confirm the broader expectation of the theory that incumbent should manipulate more extensively when they face protest threats.

Control variables The theory suggests that increasing election manipulation is indicative of regime strength, but an actor's strength—like its close analogue, power (Dahl 1957)—can only be measured relative to that of another. An incumbent who produces a high level of election manipulation may be 'strong' when facing a fragmented and unpopular opposition, but would be less so if the opposition were more formidable. As a result, it is necessary to control for potential indicators of opposition strength.

First, I include the *number of pre-election protests* counted in the ECAV dataset, which is likely to be an important possible confounder; more numerous pre-election protest may affect the level of election fraud (as the deterrence logic predicts) as well as being directly tied to the likelihood of post-election protests. Next, I control for the *median size of protest*, calculated using ECAV data.[10] This is important as a control for opposition strength, as well as for the possibility that incumbents might tolerate smaller protests. Since non-democratic governments may be able to adjust the electoral calendar strategically, to disrupt opposition coordination (Nygård 2020), I include a variable from NELDA that indicates *off-schedule elections*.

Since higher repressiveness may weaken *or* intensify opposition (Chenoweth, Perkoski, and Kang 2017), I include a lagged measure of *physical integrity* index from V-Dem (v2x_clphy), and its square term. Indicators of government *control over civil society* (the 'v2cseeorgs' variable from V-Dem), the proportion of national political parties with *permanent party organizations* (V-Dem's 'v2psorgs'), *legislative constraints* on the executive (V-Dem variable 'v2xlg_legcon', and the availability of *alternative sources of information* ('v2x_freexp_altinf' from V-Dem) are also employed as straightforward indicators of opposition strength. Since urban areas can make electoral manipulation more difficult (Larreguy, Marshall, and Querubin 2016b; Ziblatt 2009) while also making protest more likely (Wallace 2013), I also include data on *urbanization* taken from the United Nations (Department of Economic and Social Affairs (Population Division) 2019).

Other controls are included for factors that might separately cause variation in electoral

[10] ECAV data codes protest size as a categorical variable, ranging from 1 (< 10 participants) to 5 (> 10,000). I take the median size category of all the events in the protest wave as a measure of average protest size.

manipulation and protest duration. First, a variable from NELDA is used to indicate *presidential* elections, since these may include both higher levels of manipulation (Simpser 2013) and a greater risk of protest. Another NELDA variable indicates if the incumbent executive is not running due to *term limits*, which may produce elite fragmentation (Hale 2014) that affects both protest risk and electoral manipulation. De facto *judicial independence* (Linzer and Staton 2015) is included, since judicial independence can reduce manipulation (Harvey 2022a) and protest risk (Chernykh 2014). The *lagged GDP growth rate*, taken from V-Dem, is used to capture economic grievances: lower GDP growth rates, especially negative ones, should be expected to increase citizens' sense of grievance (increasing protest risk, but potentially deterring manipulation under the regime-weakness framework).

In addition, the literature on the diffusion of electoral protest holds that successful electoral protest in one country increases the likelihood of electoral protest in similar countries nearby due to opposition learning (Beissinger 2007; Bunce and Wolchik 2010). Such a dynamic could plausibly increase the severity of protest in a nearby country, while also reducing the incumbent's use of fraud. I control for this using each country's geopolitical region as coded in Teorell et al (2019), in conjunction with binary variables indicating a successful protest movement from NELDA. A 'successful protest' is recorded if a protest movement resulted in either a new election being run or the incumbent stepping down. The *regional diffusion* dummy variable takes on a value of one for all countries in the same region as a successful protest, for elections within one calendar year of the successfully overturned election.[11]

Summary statistics for all of these variables are shown in the appendix. To address concerns about multicollinearity, I also present models in the appendix that exclude control variables associated with pre-election protest.

Modeling protest duration rather than initiation helps reduce concerns about endogeneity.

[11] For example, the variable takes on a positive value for elections held in the post-communist region, within one year of the September 24, 2000 election in Serbia that sparked the 'Bulldozer Revolution'. All told there are 14 successful electoral protests in the dataset: Bulgaria 1990, Burkina Faso 1997, Lesotho 1998, Cote d'Ivoire 2000 and 2010, Peru 2000, Serbia 2000, Mali 2002, Georgia 2003, Ukraine 2004, Azerbaijan 2005, Kyrgyzstan 2005, Haiti 2006, and Cameroon 2007.

Table 3: Summary statistics and variable distributions

	Unique (#)	Missing (%)	Mean	SD	Min	Median	Max
Legislative constraints	246	10	0.4	0.2	0.0	0.3	0.9
Alternative information	313	0	0.5	0.2	0.0	0.6	0.9
GDP growth rate	250	2	0.0	0.4	−0.8	0.0	5.9
Civil society openness	265	0	0.4	1.0	−2.7	0.4	2.5
Election fraud	385	1	1.0	1.0	−2.3	1.0	3.2
Presidential election	2	0	1.4	0.5	1.0	1.0	2.0
Incumbent term-limited	3	2	0.1	0.2	0.0	0.0	1.0
Physical integrity index	277	0	0.5	0.2	0.0	0.4	0.9
Urbanization	425	0	45.4	19.0	6.8	43.1	100.0
Winning vote-share	424	9	0.6	0.2	0.1	0.5	1.0
Number of pre-election protests	32	0	2.3	9.0	0.0	0.0	100.0
Off-schedule election	2	0	1.5	0.5	1.0	1.0	2.0
Regional protest diffusion	2	0	0.2	0.4	0.0	0.0	1.0
Large protest	2	0	0.0	0.1	0.0	0.0	1.0

For instance, some regime-weakness models expect that incumbents will calibrate the level of election fraud to minimize risk of protest initiation. In that case, strong incumbents would manipulate more and weak incumbents less, but neither would systematically generate protest. Unpredictable and likely statistically insignificant correlations between fraud and protest initiation would be expected. Focusing on duration partially avoids this risk. Once protests begin (whether instrumental or multi-purpose), authoritarian incumbents have strong incentives to defeat a public challenge to their legitimacy, while committed opponents have similar incentives to stand firm. Both should be expected to put significant effort into defeating / maintaining the protest movements, as a general rule, and the process by which the challenge is settled helps reveal previously hidden information about each side; it is possible to observe whether incumbents who produce extensive election fraud are stronger or weaker than those that do not.

While strategic endogeneity is less of a concern for this research design, selection bias and confounding variables may still be. For instance, it could be the case that incumbents who face well-organized opposition-oriented civil society may manipulate less and thus provoke less protest, or even lose the election outright. Only the strongest incumbents would then 'select in' to the risk of protest by manipulating extensively, while the well-organized opposition may be able to sustain protest movements longer. A scenario like this could produce the relationship expected in H2, but would likely be considered spurious. While, as with all observational studies, it is possible that unknown or unobservable variables may introduce bias, I take several steps to reduce this risk.

First, I control for measures of opposition strength and grievances, to help isolate the relationship between manipulation severity and protest duration. Next, I develop two analyses in the appendix to assess the risk of selection bias. These analyses show that increased risk of protest leads incumbents to *intensify* their manipulation efforts, rather than diminish them; incumbents do not appear to be selecting out of manipulation when they face higher risks. Third, and relatedly, the research design pits two incompatible hypotheses against one another, rather than a single hypothesis against the null. A null relationship between fraud and protest duration might not be strong evidence against the regime-weakness model, but a negative relationship should be (after accounting for necessary controls).

5.1 Results

The large majority of elections experience no post-election protest: in 448 of the 659 elections, no protests are recorded. Figure 1 reports the distribution of election fraud across the dataset, as well as protest duration. As the figure shows, while election fraud is common among this set of regimes, protests are rare, and, when they do occur, are usually short in duration. These distributions—extensive election fraud combined with large numbers of non-protest—provides some early indication that the regime-weakness model has limited support. Still, approximately one-third of elections experienced some protest.

Table 4 reports the results of Cox proportional hazard models of protest duration. In these models, positive coefficients indicate factors that are associated with shorter-lived protests; those with a negative sign indicate factors associated longer-lasting protest. Recall that these models include data only from electoral protests that *did not succeed* in ousting incumbents.

Model 1 is a baseline with minimal controls. Models 2 and 4 do not include interactions, while the remaining models interact *intentional irregularities* with incumbent vote-share. One threat to inference in this design derives from the 90-day post-election window considered by ECAV. It is possible that some protest movements considered ‘dead’ in Models 1, 2, and 3 are, in fact, ‘alive,’ if subsequent protests occurred after 90 days and are unobserved. To address this, observations are assumed to be right-censored if a protest occurred within three-weeks of the 90-day cutoff in Models 4, 5, and 6. This amounts to 15 of the observations in the sample. Model 6 is the most restrictive model, which includes right censoring as well as a control variable for the median number of participants during the protest wave. There is considerable missingness in this variable.

The coefficient for *election fraud* is positive and significant in Models 1 and 2, indicating that more fraudulent elections are associated with less resilient protest movements. Exponentiation of the coefficient for election fraud gives a hazard ratio of 1.3, indicating that for each one-unit increase in fraud, a protest movement becomes thirty percent more likely to end at any given time. This is inconsistent with H1, suggesting that increasing election fraud is not indicative of regime weakness.

In Model 3, incumbent vote-share is interacted with *election fraud*. Figure 2 shows the predicted hazard ratios. As shown in the figure, for all incumbent vote-shares near and below fifty percent, large-scale fraud significantly increases the likelihood that a protest movement will end. This result is in line with H2 and the regime-strength model, and shows that only as vote-share increases above fifty percent—creating a stronger signal of popularity—does a government in a low-fraud election have the same likelihood of defeating a protest movement at a given time.

The results of the baseline models are consistent with those models which incorporate censoring. The coefficient for *election fraud* retains the same size, sign, and level of significance as in Model 2. The interaction effect in Model 5 increases in size, while retaining its direction and statistical significance. This suggests that the main models are not drawing overly incorrect inferences through censoring of longer-lasting protest movements. Model 6, adds the median number of participants at events in the protest wave as a control for the potential perceived threat of the protest movement by the incumbent. Including this control variable reduces the sample size due to missingness, and standard errors increase as a result. Nevertheless, the coefficients on election fraud and the interaction term retain their size and direction.

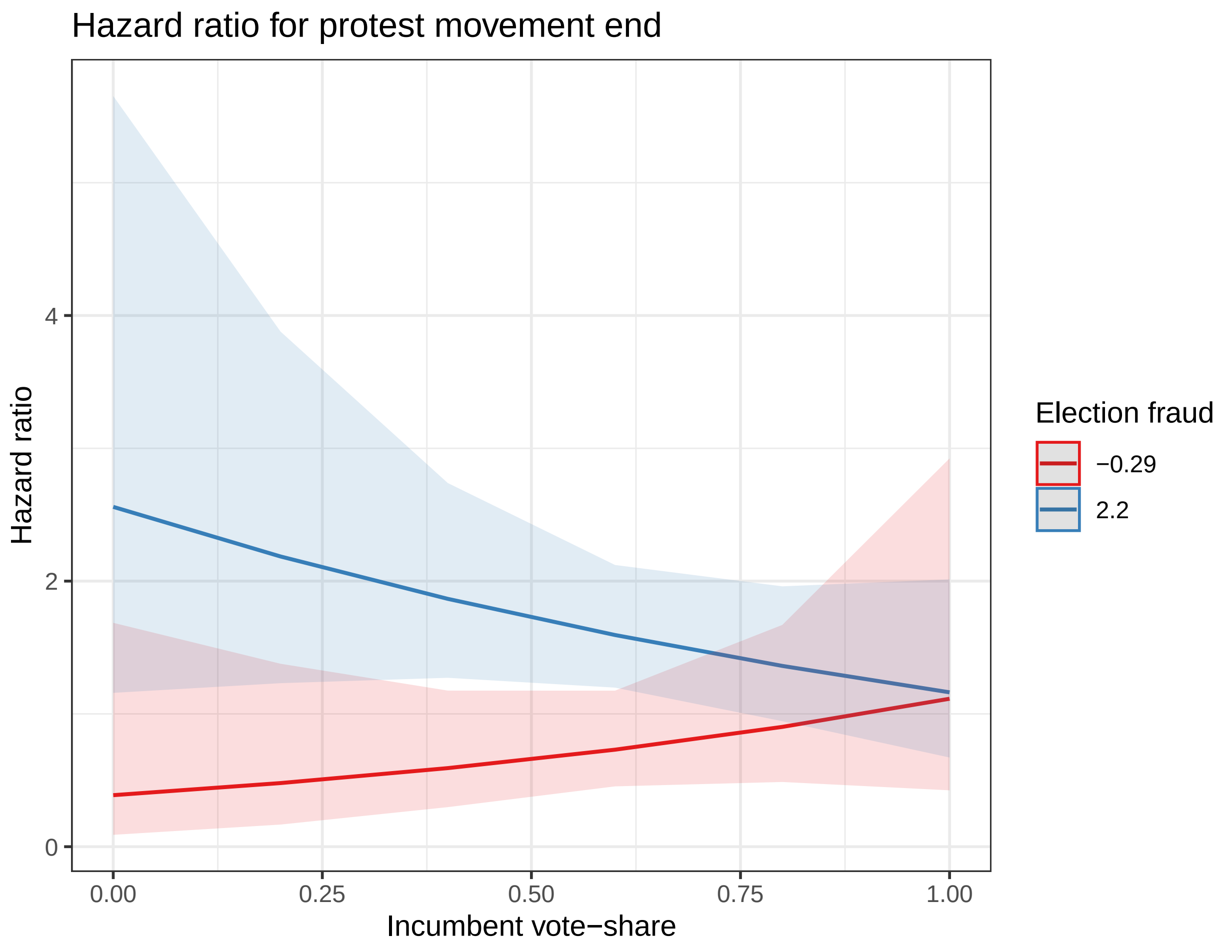

Figure 2: Fraud, winning vote-share, and duration of protest

The relationship between the level of fraud and the survival of protest movements can also be demonstrated non-parametrically using Kaplan-Meier curves for varying levels of fraud. In Figure 3, observations are divided into three groups. High fraud observations are those where *election fraud* is one standard deviation above the mean or more, low fraud observations are those one standard deviation below the mean or less, with moderate levels in between. The figure shows a clear difference between the low-fraud group and the other two. Low-fraud cases are much more likely to see their protest movements survive past the first day than their counterparts. For much of the range of the data, they remain roughly twice as likely to survive at any given date than protest movements in medium- or high-fraud cases. This is evidence in favor of the idea that high levels of fraud indicate stronger regimes.

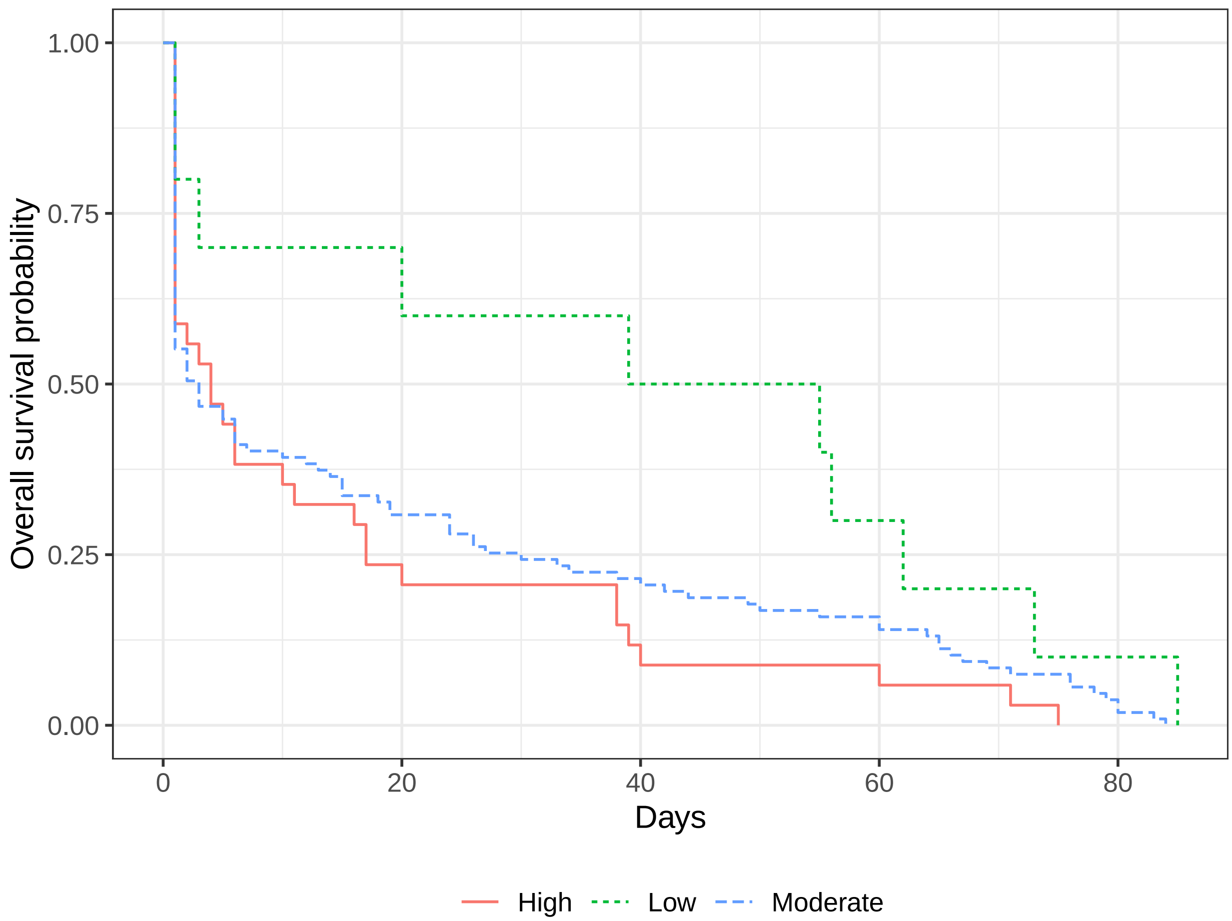

Figure 3: Kaplan-Meier curves for protest duration across levels of fraud

6 Discussion

The duration analysis consistently shows that more election manipulation is associated with quicker incumbent victories over protest movements, both independently and in conjunction with low incumbent vote-share. In real contests in which opposition and incumbent strengths

Table 4: Table 4: Cox proportional hazard models of protest duration

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Judical independence (lag)		−0.14	−0.01	0.09	0.21	0.04
		(0.93)	(0.94)	(0.98)	(0.98)	(1.26)
Leg. constraints (lag)		−0.55	−0.48	−0.78	−0.69	−1.01
		(0.55)	(0.55)	(0.59)	(0.58)	(0.84)
GDP growth rate (lag)		0.14	0.15	0.15	0.16	0.97
		(0.18)	(0.18)	(0.18)	(0.18)	(1.83)
Alternative info. (lag)		0.79	0.90	1.00	1.22	0.62
		(0.91)	(0.92)	(0.96)	(0.97)	(1.41)
Civil soc. openness (lag)		−0.23	−0.25	−0.16	−0.20	−0.07
		(0.18)	(0.18)	(0.19)	(0.19)	(0.24)
Presidential election	−0.18	−0.08	−0.08	−0.14	−0.14	−0.10
	(0.18)	(0.21)	(0.21)	(0.22)	(0.22)	(0.31)
Incumbent term-limited		−0.40	−0.41	−0.33	−0.36	−0.57
		(0.34)	(0.34)	(0.36)	(0.36)	(0.44)
Urbanization		0.002	0.002	−0.0006	−0.001	−0.002
		(0.006)	(0.006)	(0.006)	(0.006)	(0.009)
N. of post-election protests	−0.05***	−0.05***	−0.05***	−0.07***	−0.07***	−0.05***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)
Off-schedule election		0.34	0.32	0.36	0.32	0.35
		(0.23)	(0.23)	(0.23)	(0.23)	(0.33)
Phys. integrity index (lag)		1.02	0.36	0.87	0.09	1.54
		(2.04)	(2.09)	(2.13)	(2.16)	(2.74)
Phys. int. lag squared		−0.68	0.02	−0.60	0.26	−1.26
		(2.18)	(2.23)	(2.23)	(2.27)	(2.84)
Regional protest diffusion		−0.14	−0.18	−0.04	−0.12	−0.22
		(0.21)	(0.21)	(0.22)	(0.22)	(0.27)
Median protest size						0.04
						(0.14)
Incumbent vote-share	1.19	−0.34	0.84	−0.40	1.24	0.77
	(0.88)	(0.54)	(1.00)	(0.55)	(1.03)	(1.60)
Election fraud	0.82**	0.29*	0.76*	0.34*	0.97**	0.88+
	(0.32)	(0.14)	(0.37)	(0.15)	(0.38)	(0.53)
Election fraud:Inc. vote-share			−0.74		−1.02+	−0.94
			(0.53)		(0.55)	(0.78)
Num.Obs.	143	135	135	135	135	93
RMSE	0.99	0.95	0.95	0.89	0.89	0.88

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

are tested against each other, better manipulators reveal themselves to be stronger rather than weaker. This holds true even when incumbent vote-share is low, when the regime-weakness model would make the opposite prediction—in that model, incumbents who are unpopular and manipulate heavily ought to be the weakest and struggle hardest to quell protest. It is therefore inconsistent with Hypothesis 1, casting doubt on the 'regime-weakness' model of manipulation. Further, the relationship holds despite controlling for indicators of other possible explanations of the relationship between low fraud and long-lived protest, including civil society strength, media openness, regional electoral protest waves, number of protests and median protest size. While this study does not purport to show perfect causal identification, these robust correlations are presented as incompatible with the regime-weakness framework and supportive of the regime-strength model of manipulation.

Specifically, the results highlight patterns that might be considered puzzling, whichever model of electoral manipulation we prefer. For the regime-weakness model, it is potentially puzzling that most manipulated elections go un-protested. Regime-weakness models may explain this pattern by arguing that, most of the time, incumbents produce the 'right' amount of manipulation to avoid protest in equilibrium (Little 2012, e.g.). Such careful calibration is difficult to ensure, however, when incumbents must rely on thousands of low-level agents to produce the 'right' amount of manipulation; these agents may have collective incentives to shirk and underproduce (Rundlett and Svolik 2016), or to demonstrate their loyalty by over-producing (Cho and Park 2025). The results of this study add an empirical challenge, since it demonstrates that when the deterrence equilibrium fails (i.e. when the incumbent rigs elections and the opposition protests), higher levels of manipulation are associated with greater regime strength, not less.

In rejoinder, it might reasonably be wondered why—if manipulation indicates strength—why protests occur after highly manipulated elections? In my view, this interpretation unnecessarily exports the instrumental logic of protest from the regime-weakness model to the regime-strength model. As discussed earlier in the paper, instrumental calculations about overthrowing incumbents are but one component of a long and varied literature on protest, which has identified multiple other expressive and strategic motives for protest. These other motives may foster protest movements in the aftermath of rigged elections, even when severe electoral manipulation indicates considerable regime strength. Second, if manipulation does indeed indicate strength, incumbents have every incentive to build and deploy those strengths in reaction to stronger opposition movements (who will themselves be more capable of initiating protest). Evidence that incumbents do, indeed, appear to increase manipulation in the face of increased protest threat is presented in the supplementary appendix.

It also helps us make sense of secular patterns that are puzzling if election fraud indicates regime weakness, as others have held. The pattern depicted in Figure 1 for example, which shows high levels of manipulation and low levels of protest, is sensible in a world where the more powerful and durable incumbents produce greater levels of manipulation, and where incumbents who struggle to produce high levels of manipulation are ones who face resource scarcity, institutional barriers to impunity for their agents, and defections among erstwhile allies. It also helps explain why some incumbents manufacture their way to highly implausible margins of victory, reasoning that a demonstration of their extra-electoral power may suggest to allies and bystanders that the incumbent may survive protest despite a low natural popularity. Scenarios where incumbents use massive fraud to post margins of victory in the eighty and ninety percentage point range are harder to explain under a worldview where more manipulation indicates deeper regime weakness.

What does all this tell us? First, these findings support and extend the line of research initiated by Simpser (Simpser 2013). Understanding election manipulation as indicative of effective sources of regime strength comports with a growing literature on the principal-agent dynamics of manipulation (Harvey 2019, 2022a; Rundlett and Svolik 2016) and the importance of patronage resources and electoral processes in sustaining authoritarian regimes (Hale 2014). This study offers empirical evidence that authoritarian governments who manipulate more are stronger than those who manipulate less (all else equal). It further illustrates how the mechanisms which underpin Simpser's model of manipulation-as-strength—coordination effects among pro-regime agents and costly signaling of incumbent resources—can function even when such signals of strength have failed to 'deter' protest.

Second, these results may inform how researchers think about the causes of election manipulation. They suggest that the degree to which the opposition's capacity to organize street protest can limit election fraud has been overestimated. Consequently, theories of electoral manipulation that emphasize alternatives—such as resource availability (Greene 2007), principal-agent problems (Rundlett and Svolik 2016), and legal risks to manipulators (Harvey 2022a)—should be further developed, and may yield new policy interventions. In particular, it may change some current understandings of the role of election monitors, courts, and other third parties in upholding election integrity. As noted earlier, prior studies argue that these actors restrain manipulation by revealing information about fraud and intensifying protest risk. However, if more extensive manipulation is associated with more quickly subduing protest movements, it may be that other mechanisms are at work. For instance, it may be that election observation makes it harder for front-line agents to avoid personal consequences for engaging in manipulation (Harvey 2022a).

By showing that the infrastructure of fraud can actually reduce the cost of post-election protest to the regime, this result poses a challenge to theory of democratization by election (Edgell et al. 2018; Morgenbesser and Pepinsky 2019; Schedler 2013). If the capacity to engage in widespread manipulation makes it easier to both tolerate the existence of an opposition and to suppress it during periods of contestation (Dahl 1971), electoral authoritarianism may be something of an institutional cul-de-sac: the capacity to generate severe election fraud makes it appealing for closed authoritarian regimes to introduce elections, while the manageable risk of protest makes it unappealing to further democratize. Likewise, the results suggest that, on average in authoritarian regimes, the benefits of heavy manipulation outweigh the legitimacy costs incurred by incumbents for such malfeasance (Birch 2011). Future research might investigate the scope conditions for this pattern; cases where the legitimacy costs of fraud are very high may prove to be instances where the ‘regime-weakness’ model is more appropriate.

Finally, this paper does not examine the role of protest risk in insulating established democracies from democratic backsliding (Fearon 2011). This question deserves further investigation, but it is possible to draw a few implications from this study. First, even if protest risk fails to serve as a deterrent in non-democracies, it may still do so in more democratic settings. The likelihood of protest initiation in a democracy may well be higher, triggering at lower levels of fraud and higher incumbent vote-shares, since opposition actors are likely to be better resourced in a liberal democracy (and the government more constrained). At the same time, if the theory described here is accurate, liberal democracies would be constrained against using large-scale election fraud even in the absence of protest risk: they lack the patronage resources and arbitrary enforcement of laws needed for such a scenario.

These results remain consistent under a several alternate specifications presented in an online appendix. The results are supported using two alternative measures of election manipulation, one that covers a broader array of electoral violations, and another that is drawn directly from election observer reports. Additional tests show that incumbents intensify their efforts to manipulate when protest risk is high, that higher capacity for manipulation makes incumbent election loss less likely, and that the small number of successful protests occur across the spectrum of manipulation. This helps rule out an interpretation of the main results in which weaker regimes choose not to manipulate and thus avoid protest. The results also remain consistent when the selection criteria are relaxed, and a set of electoral democracies are included in the dataset.

7 Conclusion

There are competing views on the relationship between election fraud and protest. In one family of models, which I have called the regime-weakness model, incumbents may hold cleaner elections when the risk of post-election protest is higher. Other models contend that election fraud can indicate incumbent strength. Building on the principal-agent literature of election fraud, I argue that large-scale fraud draws on the same resources that make incumbents resilient to protest, even when regime popularity (and thus vote-shares) are low. Conducting election fraud is made easier when incumbents have control over patronage resources and the ability to protect front-line agents from the consequences of illegal behavior; this capacity also helps insulate incumbents from the effects of mass protest. Resolving the tension between these views in the literature has important implications for the study of election fraud and authoritarian stability.

This paper tests the assumptions of these two models by investigating the relationship between election fraud and the duration of protest. Though both models may make the same predictions regarding protest onset; their predictions differ with regard to protest duration. Drawing on data from the V-Dem, NELDA, and ECAV datasets, it uses cross-national data on 659 elections in electoral authoritarian regimes to show that incumbents that engage in larger-scale election fraud are able to more quickly defeat protest movements.

All together, these results suggest that protest risk is unlikely to serve as a constraint on election manipulation in authoritarian regimes in the way that the regime-weakness model predicts—the most effective election manipulators are also the most well-prepared to defeat protest movements. The frailty of protest risk as a guardrail suggests that electoral authoritarianism as a regime-type is a stubborn equilibrium. This is a gloomy picture for supporters of democratization, and suggests that researchers should further consider other mechanisms beyond mass protest in their models of electoral manipulation.

8 Data availability statement

The data analyzed in this study will be made available at the Dataverse upon acceptance of the paper.

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